Real-time Grasp Affordance Detection of Unknown Object for Robot-Human Interaction

Azhar Aulia Saputra, Chin Wei Hong, and Naoyuki Kubota

***Abstract*— By using a combination of vision and depth map sensors, this paper aims at detecting the real-time affordability of gripping pose for hand-over object behavior in robot-human interaction. The affordance detection will serve a set of seven-dimension gripping information (3D location, 3D Rotation, and gripping size). Moreover, the novelty is that the goal has to consider gripping behavior of the receiver. Therefore, the result of the proposed method discusses the gripping location of the robot and the estimation of the receivers gripping location. Technically, desired objects detection is computed using a computer vision algorithm. After that, from the depth information, the topological map will be generated using the proposed dynamic density growing neural gas. The density topological structure will be focusing on the desired object. With the topological map information and robot gripper embodiment, the possible gripping position is computed based on the inlier-outlier method. Ranking information in every detected gripping is also considered for classifying from the best and the worst gripping position. Experimental results showed that the proposed work capable of detecting the affordance with a gripping recommendation in real-time with a low computational cost.**

***Index Terms*— Grasped Affordance detection, Topological map building, Object detection**

I. Introduction

Many researchers have dealt with robot grasping behavior. There is a lot of invention model built, and also many challenging problems have been done. However, there are still many problems in space that should be solved. One of the challenging problems is the speed of gripping decision. The targeted object's position can change due to a specific reason. Therefore the decision process should be real-time if it deals with the uncertain position of the object. As an example, when the human hand-over certain object naturally, the given object posture cannot be exactly static. Still, there are some object movement should be considered by the receiver when he/she is grasping it. Furthermore, there are many cases in disaster condition should consider, such as the object movement due to the earthquake, grabbing the roped ladder, etc. The essence is that the algorithm has to keep paying attention to the object condition.

There are a lot of model or algorithm that able to detect the gripping pose under noisy data with many variations in the environmental condition. Some algorithms also dealt with unknown object detection [1]. However, the learning process of gripping decision takes some time. It is not processed in real time. It will be no problem when it dealt with static object posture. When there is a changing posture, it will grasp the object incompletely. We therefore in this paper propose a fast and efficient strategy and learning algorithm for detecting affordance of gripping. The proposed algorithm can detect real-time a set of gripping position possibilities.

Vision-based recognition is the most straightforward way for hardware installation and configuration. However, it required a robust learning model such as CNN or deep learning that required a high computational cost. Lightening problem may cause the exposure and brightness that disturb the recognition process. Therefore, our proposed model combines two sensory information which is vision and depth sensor. The depth sensor will cover the vision problem with the depth information. However, they are separately processed. The vision part aims to detect the desired object. It will share the frame information representing the detected object.

We developed an array depth sensor, a combination of four time-of-flight sensors. It generates around 200 thousand point information. Therefore, It will be costly for processing without data reduction. In other hands, a topological structure can represent the raw data with lower data information. However, the topological structure may not be able to clarify small things. It unclear represents the object that has much texture. Therefore we proposed an algorithm to improve the performance of topological map generation. It can generate with dynamic density. The density of topological structure represents the targeted object is much increased, so that it can clarify well.

We use the topological information and robot gripper as the considered input data. We use inlier and outlier mechanism for recognizing the possible gripping position. There are a set of possible gripping positions will be served with a ranking mechanism calculated by proposed fitness evaluation.

In human-robot interaction, the robot should not only grab an object but also hand over the object to the receiver. In order to perform naturally, the robot has to imitate human behavior. Cini et al. proposed human behavior when grasping some object and pass it to someone [2]. Human tends to understand the purpose of the object and give the handler to the receiver [3]. As the example, when we pass a knife for use by the receiver, we leave the knife handler unobstructed to facilitate the receiver can hold comfortably. In order to aim that principle, we improve our proposed model therefore able to estimate the most comfortable position for the receiver and recommend a gripping position for the robot as the passer. In the final result, our algorithm will serve the affordance information for grasping that composes the possible grasping

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position with a degree of safe graspable. Furthermore, two recommendation grasping position will be generated for passer and receiver gripping position.

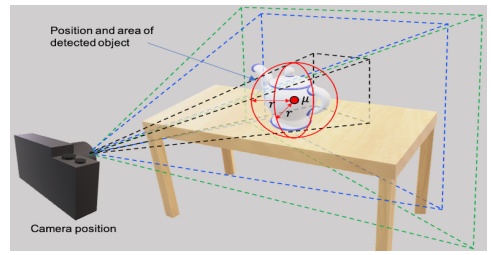
II. Related Work

The development of image processing has been increased significantly. It can recognize moving objects [4], with real-time detection [5]. Currently, researchers implement deep learning model for solving difficult problems [6], [7]. In grasping position detection, Some researchers estimate the object pose in order to decide the grasping location using deep learning. However, when only using vision information, it requires high computational cost with limited result. Therefore, if there is a possibility, some researcher tends to combine with depth information for recognizing the pose of the object [8], [9]. Pas et al. used 3D point cloud information generated from a RGB-D camera for detecting the object grasping pose [10].

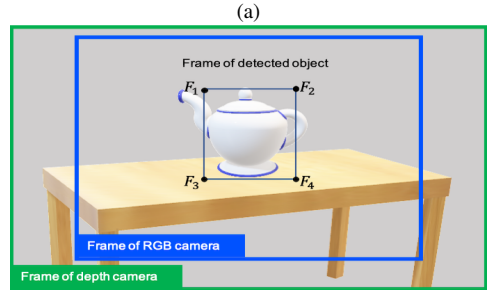
Some researchers use RGB-D camera [8], [9], [11] for detecting grasping position. Deep learning is also a popular learning model in grasping position detection [11]. Kumra et al. implemented it and resulted in 89.21 % successful rate on the standard Cornell Grasp Dataset [11]. Lenz et al. [8] use a deep neural network. He proposed fast and robust detection by proposing two cascades deep-learning model. The first layer of deep learning has fast computation conducting in short term adaptation and the second layer has higher computational cost processing in long term adaptation. They implemented in visual servoing in Dexter robot grasping. The proposed model in [9] also use a two-step learning process with different term adaptation. The model generates 7-Dimension information (3D position, 3D rotational information, and size of grasping area). However, their implementation did not show the effectiveness in moving object and real-time detection. Redmon et al. proposed real-time object grasping detection using convolutional neural network [5]. It takes a preliminary learning process and takes a lot of data collection. Therefore, they did not deal with unknown object detection.

The deep learning usage needs to take the long training process and requires a lot of data collection [12], [13]. In grasping detection case, it is difficult to deal with unknown targeted object. Some researcher using the geometric analysis for detecting the grasping position in unknown object [14], [15]. Sahar et al. proposed a strategy called the Grasping By Components or GBC [14]. They use object segmentation, shape approximation, and learning process for detecting graspable component. While, other researchers use only vision information for detecting unknown object [16]. However, it takes a supervised learning process, and their implementation does not show the moving object. In order to deal with an unknown moving object, we analyze the geometry of the object's components using topological map information with a low computational cost.

There are many algorithms for unsupervised learning model for 3D point cloud data [17], [18], [19]. We developed topological map building based on Adaptive resonance theory (ART) [20], [21]. Some researcher has built growing neural gas (GNG) that shows its effectiveness to generate a topological map from 3D-Point Cloud data [22]. It has the same density of nodes in every part of the topological map. Therefore, when a small object put together with the big object, the topological map will be difficult to clarify the shape of the small object. Therefore, we propose a dynamic density of topological map building. Our model can clarify the shape of the targeted object represented by the topological structure. Implementation on grasping robot, SVM has been used for learning optimal grasping position and implemented on grasping robot [23].



(a)



(b)

Fig. 1. The illustration for generating the object location and area (a) Perspective view of object detection (b) Frame view shows the four frame point of object detection and comparison of depth and RGB camera's frame.

III. Grasping Affordance Detection

The proposed algorithm aims to detect the affordance of grasping. The affordances serve seven dimension grasping position. It uses a combination of RGB camera and depth camera as the sensory information. RGB camera is used for detecting the desired object and Depth camera for detecting the affordance of grasping. However, the object detection has been done by Redmon et al. [5]. We use their YOLO v3 Framework [24] for object detection.

The detected frame position and size served by object detection part will be matched to the depth camera frame and position. After that, we calculate the average position (μ) of the depth points in the selected frame ($\mathbb{X}$) and calculate the radius of the detected area by calculating as $r = 2 * \sigma$, where σ is the standard deviation of $\mathbb{X}$. The illustration can be seen in Fig. 1. The spherical area information will be used for the dynamic density topological map building. The main algorithm of our proposed grasping affordance detection can be seen in Algorithm 1. In the first state, object detection is conducted. After that, it generates the topological map using dynamic density growing neural gas. Then, the grasping affordance detecting will be processed.

Algorithm 1 Dynamic Topological Structure

Initialization
loop
  $\mu, r \leftarrow$object detection framework ()
  Generate dynamic density topological structure ()
  //grasping affordance detection
  for $i \leftarrow 1$ to $\max_\lambda$ **do**
    Generate new possible grasping position
    Evaluating new possible grasping position
    Update and evaluate current grasping position
    Calculate the rank of grasping position

A. Dynamic Density Topological Map Building

The topological map building aims to decrease the number of data information by representing all raw data input with several points and edge information. We used GNG based model with dynamic density. It can be more clarifying small and detail object than other common GNG. The previous development of our GNG has been implemented for real-time topological map building [22]. In this model, we add a parameter in every node for representing the strength of the node. The strength parameter of kth node (δ_k) will be calculated in Eq. (1),

$$\delta_k = \frac{e^{-(r+\|h_k-\mu\|)}}{1+e^{-(r+\|h_k-\mu\|)}} \tag{1}$$

where h_k, r, and μ are the kth 3D node's position, the radius of sphere, and 3D position of sphere.

Data collection ($\mathbb{V}$)in the proposed DD-GNG is also affected by the spherical area generated by object detection. The data will be randomly collected with different search space area. When t is an odd value, the data will be generated from all raw data ($\mathbb{P}$). When t is an even value, the search space is only inside the sphere(r,μ) area. The data collection of DD-GNG can be seen in the Algorithm 2.

Algorithm 2 Dynamic Topological Structure

Init: generate two nodes at random position
  $c_{i,j} = 0$ $(\forall_i, \forall_j)$, $A = 1,2$, $r = 2, t = 0$
loop
  $\mathbb{P} \leftarrow$ raw data from depth sensors
  $\mathbb{Q} \leftarrow$ raw data inside $Sphere(r,\mu)$
  $\lambda \leftarrow 0$
  for $i \leftarrow 1$ to $\max_\lambda$ **do**
    if $t \mod 2 = 0 \bigwedge$ object detected **then**
      $V_\lambda \leftarrow$ a random of $\mathbb{Q}$
    else
      $V_\lambda \leftarrow$ a random of $\mathbb{P}$
    $\lambda \leftarrow \lambda + 1$
  DD-GNG main process (V)
  Calculate discount rate ()
  $t \leftarrow t+1$

Furthermore, the strength parameter will effect to utility rate calculation. The utility value of vital nodes will be slower decreased than the utility value in weak nodes. The calculation can be seen in Algorithm 3, where E_i and U_i are the GNG error value and GNG utility value of ith nodes. The node with the utility value lower than the defined threshold is removed from the topological structure, $E_u/U_l > k$. The utility value will effect to the node deletion and node addition. A weak node will easy to be deleted and strength node will easily produce a neighbor node.

Algorithm 3 Calculate discount rate

for $i \leftarrow 1$ to node's number **do**
  $E_i \leftarrow E_i \times \upsilon_E$
  $U_i \leftarrow U_I \times \upsilon_E / \delta_i^2$

B. Detection of Grasping Position

The proposed detection of grasping detection uses the topological structure generated by DD-GNG as the input data rather than the raw data. It will only consider the vital nodes with the strength value bigger than the defined threshold. The proposed detection will detect multiple possible grasping position by considering the gripper embodiment. The gripper embodiment composed as the tightness of the gripper, maximum and minimum gripping can be freely defined.

1) Generating Possible Grasping Position: In order to find the perspective grasping position, two strong nodes (h_{n1}, h_{n2}) will be randomly generated. After that, the nodes will be evaluated by considering the edge connectivity and the density ratio of inlier and outlier nodes surrounding the pivot from n_1th and n_2th nodes. We calculate the evaluation value as computed as Eqs. (2) and (3), where c_i and c_o are the number of strong nodes in the inlier and outliers area respectively. In Eq. (3), when the evaluation value higher than the threshold λ_Θ, then the generated possible gripping position will be added. Otherwise, the detected will be deleted. The illustration of accepted or rejected generation is shown in Figs. 2. The node detection and the length of the pivot should not longer the the tightness of gripper (l_{max}), see Fig. 2a.

$$\Theta = \frac{(c_i)/(1+c_o)}{\|\mathbf{n}^{(2)} - \mathbf{n}^{(1)}\|} \tag{2}$$

$$\text{Eval. result} = \begin{cases} \text{Accepted, } N_g++ & \text{if } \Theta \geq \lambda_\Theta \\ \text{Rejected} & \text{otherwise} \end{cases} \tag{3}$$

2) Updating Grasping Position: This process aims to correct the axle position of grasping $\bar{\mathbf{n}}$ in the center position of node's members. The illustration of this process and the parameters used can be seen in Fig. 3. We will consider and calculate only the nodes surrounding the axle of the grasping position. First, we will calculate the similarity of the nodes using Eq. (4),

$$s_{i,k} = \frac{\|\mathbf{v}_i - \mathbf{v}_k\| \cdot |\theta_i - \theta_k|}{\|\mathbf{d}_i - \mathbf{d}_k\|} \tag{4}$$

where v_i and d_i is the parallel and perpendicular distance between ith node and the axle. (see Figs. 3a and 2a).

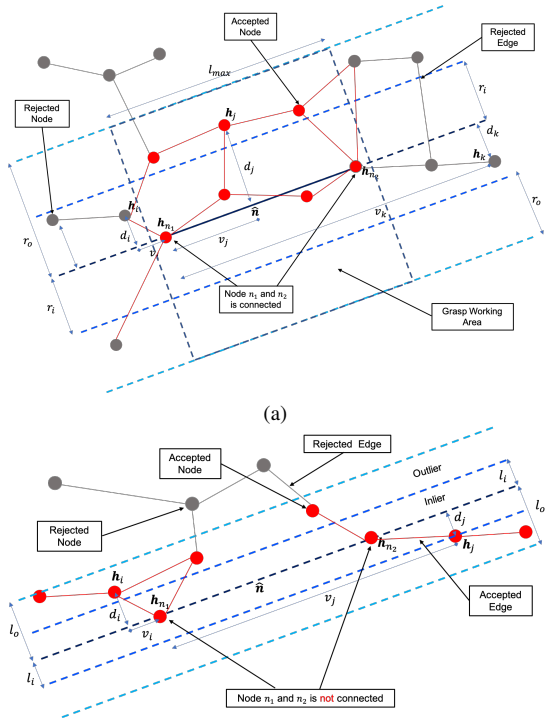


Fig. 2. Illustration of topological structure (a) Generated nodes n_1 and n_2 is considered as a candidate of possible grasping position. The figure show the parameter involved for evaluation and position correction. The parameter l_{max} represents the tightness of the gripper. (b) Illustration of evaluation when generated nodes n_1 and n_2 is not connected. It is not considered candidate.

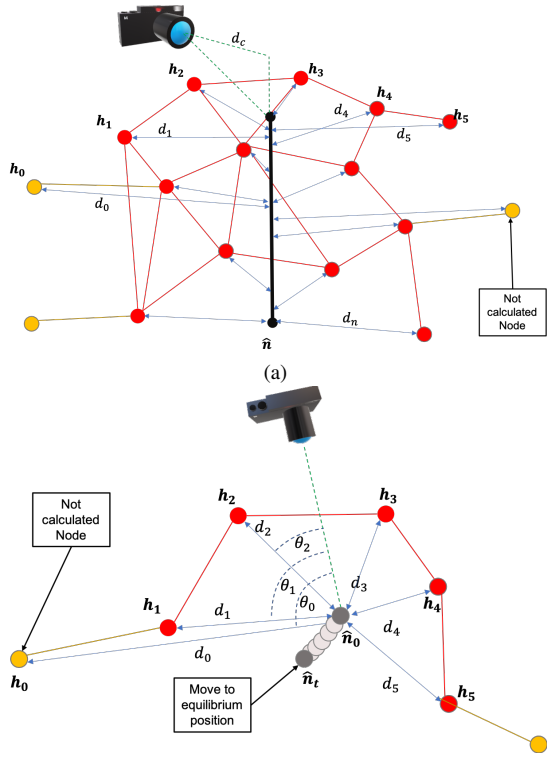


Fig. 3. The illustration for generating the object location and area (a) Perspective view of object detection (b) Frame view shows the four frame point of object detection and comparison of depth and RGB camera's frame.

Parameter $s_{i,k}$ represents the degree of similarity between ith and kth nodes. If there are same nodes, then we will only consider the nearest nodes.

The alteration movement of starting axle node and end axle node are calculated in Eqs. 5 and 6, where c_i is the number of node in inlier area.

$$\Delta e_1 = \frac{1}{c_i} \sum_{j=\mathfrak{R}\to N_i}^{j=0} \frac{\|\mathbf{v}_j\|}{\|\mathbf{v}^{(2)}\|} (\mathbf{r}_j - \mathbf{v}_j) \tag{5}$$

$$\Delta e_2 = \frac{1}{c_i} \sum_{j=\mathfrak{R}\to N_i}^{j=0} \frac{\|(\mathbf{v}^{(2)} - \mathbf{v}^{(1)}) - \mathbf{v}_j\|}{\|\mathbf{v}^{(2)}\|} (\mathbf{r}_j - \mathbf{v}_j) \tag{6}$$

N_i is the number of nodes involve in ith detected rung. After calculating the error position of detected grasping position, the start and end points are updated using Eqs. 7 and 6.

$$\mathfrak{R} \to h_1 = (\mathfrak{N} \to \mathbf{R}_{h_1} - \mathbf{v}^{(1)}) - \varepsilon \Delta e_1 \tag{7}$$

$$\mathfrak{R} \to h_2 = (\mathfrak{N} \to \mathbf{R}_{h_1} - \mathbf{v}^{(2)}) - \varepsilon \Delta e_2 \tag{8}$$

h_1 and h_2 are the start and end point of the detected rung.

3) Calculating grasping direction: In order to get simple assuming, we consider the average vector of the associated node from the axle of grasping detection. The direction of the grasping vector of jth grasping position is calculated in Eq. 9. Parameter i indicates the id of the associated neuron. N is the number of neurons associated with a certain grasping position. The calculation will be affected by the degree of similarity to other nodes ($s_{i,k}$).

$$\theta_j = \sum_{i=0}^{i<N} \sum_{k=0}^{k<N} (s_{i,k}) \cdot \frac{\mathbf{v}_i}{N} \tag{9}$$

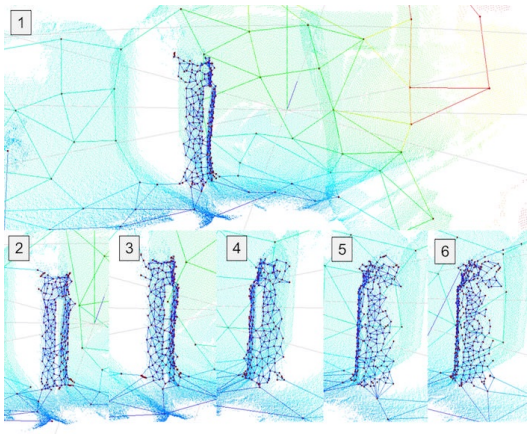
IV. Experimental Result

These experiments show the effectiveness of our proposed dynamic density topological map building combined with visual information representing the shape of the object. After that, we will show the grasping affordance detection for static object and moving object. We used standard Web camera as the visual sensor and used four time-of-flight sensors the depth sensors. Each of the sensor coverage range of $45^0 \times 62^0$. Therefore, they can cover the view of $165^0 \times 62^0$. We perform the experiments in computer NUC i5 with 8 GB of RAM without any GPU installed.

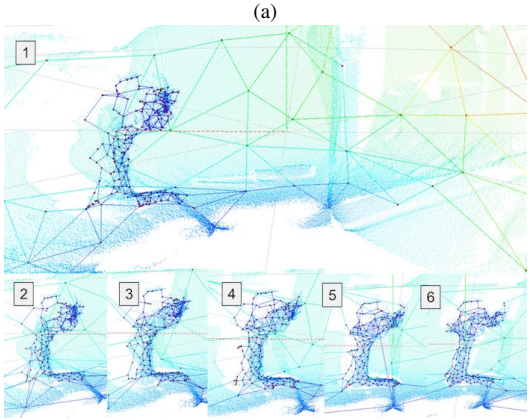
A. Dynamic Density Topological Map

Before testing the grasping affordance detection, we will test the proposed topological map building combined with object detection framework. The parameter setting of the DD-GNG is adopted from the basic idea in [22]. There are 300 points from all raw data generated randomly is processed in the proposed GNG.

There were 2 moving objects tested, a bottle and a radio remote control. The result is shown in Fig. 4. The real-time



(a)



(b)

Fig. 4. The snapshot of the proposed dynamic density topological map building in moving object. Small dots represent the raw data of the depth sensor and the black dots with edges represent the generated topological map. (a) a bottle. (b) a radio control

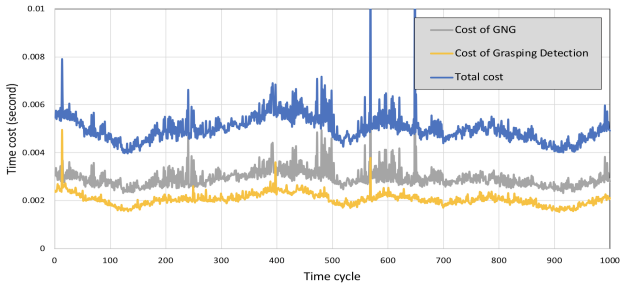
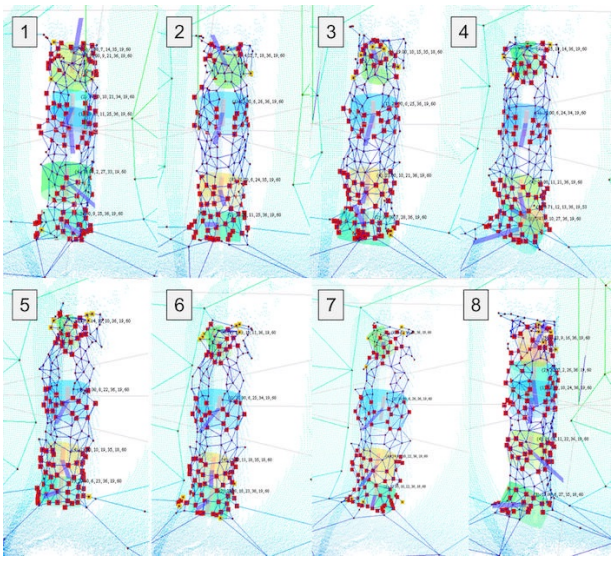


Fig. 5. The graph of time consuming during the proposed model performance

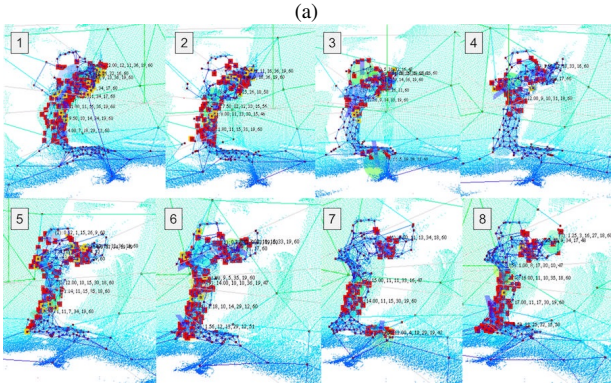
performance is achieved. By increasing the density of structure in the object area, the object's shape can be clarified. It required 0.003 seconds for executing the topological map building. Therefore, it can be used for real-time detection. However, the computational cost can increase and decrease depending on the number of node and parameter λ in the topological map.

B. Grasping Affordance Detection

In order to prove the effectiveness of the proposed model, we tested the detection model in static objects and moving objects. In these experiments, there were three objects used as the sample, which are screw-driver, radio remote control,



(a)



(b)

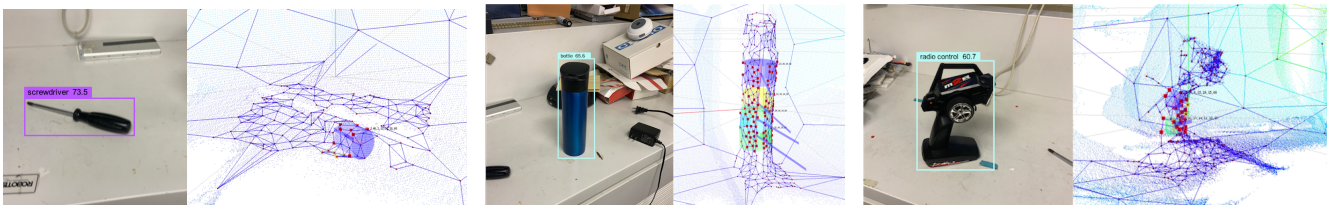
Fig. 6. The snapshot of the proposed grasping affordance detection for moving object. (a) a bottle. (b) a radio control

and a bottle. The experiments for the static object are shown in Fig. 7. There are several possible grasping positions in the radio remote control and a bottle. However, there is only one possible grasping position in the screwdriver. The size of the object influences the number of detected possible grasping position. The defined gripper size also affected. If the tightness of the gripper is small, then the possible grasping position will be generated more. The grasping affordance detection model requires around 0.002 seconds for executing. However, this can increase and decrease depending on the number of detected possible grasping position.

In order to prove the efficiency of computation, we tested the proposed model for detecting a moving object. Fig. 6 shows a snapshot of successful performances.

V. Discussion and Conclusion

We have proposed the real-time object affordance detection for robot grasping. The sensory input is composed as the combination of visual information and depth information. However, it works separately, visual information for detecting the desired object using a common object detection framework. Our proposed work mostly occupy the depth of sensory information. We propose the dynamic density



(a) (b) (c)

Fig. 7. The left figures are the result from object detection framework. The right side figures are the snapshot of the proposed grasping affordances detection for static object. There are 7 information provided every detected grasping position. 3D position of tube, dimension of tube represents the tightness of the grasping, and 3D Blue vector represents the grasping direction. (a) a screw driver (b) a bottle (c) a radio control.

topological map building for decreasing the number of depth data representation. We add strength parameter in every node and give higher value where the position is approaching the center of the selected frame detected by object detection. The novelty of the proposed model has been shown in the experiment Section IV-A. The proposed topological map generator can clarify the desired object by increasing the density of node specific on the object area. This model can solve the obscurity of a small object by using topological map building with the low computational cost. DD-GNG required around 0.003 seconds for processing the data. Therefore, this model can be used for other implementation instead of object grasping detection for achieving lower computational cost.

In this paper, the topological map generated by proposed DD-GNG is used for grasping detection. The affordance detection will serve a set of seven-dimension gripping information (3D location, 3D Rotation, and gripping size). The experiment in section IV-B shows the effectiveness of the proposed model detecting the grasping affordances of the desired object. There are several possible grasping positions, each grasping position is represented by a tube with an arrow. The tube has 4 information which is the 3D position, and the diameter represents the tightness of the grasping. The arrow represents the posture direction of the grasping. It also succeeded in performing detection for moving object. Comparing with current grasping detection, the proposed model has a lower computational cost. Therefore it can detect real-time moving objects.

Since this experiment only is limited to the affordance detection of the grasping, in the future, we will conduct handover behavior experiments. Since the cognitive model has not been integrated, we will decide the grasping recommendation for giver and receiver based on the ranking parameter given by the proposed model. After that, the learning model will be built for detecting and analyzing a novel object. Thus, the appropriate grasping position can be detected based on the result of object behavior analysis.

REFERENCES

[1] A. Saxena *et al.*, "Robotic grasping of novel objects using vision," *The International Journal of Robotics Research*, vol. 27, no. 2, pp. 157–173, 2008.

[2] F. Cini *et al.*, "On the choice of grasp type and location when handing over an object," *Science Robotics*, vol. 4, no. 27, p. eaau9757, 2019.

[3] K. Strabala *et al.*, "Toward seamless human-robot handovers," *Journal of Human-Robot Interaction*, vol. 2, no. 1, pp. 112–132, 2013.

[4] H. Cai, Q. Wu, T. Corradi, and P. Hall, "The cross-depiction problem: Computer vision algorithms for recognising objects in artwork and in photographs," *arXiv preprint arXiv:1505.00110*, 2015.

[5] J. Redmon and A. Angelova, "Real-time grasp detection using convolutional neural networks," in *IEEE Intl. Conf. on Robotics and Automation (ICRA)*. IEEE, 2015, pp. 1316–1322.

[6] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in neural information processing systems*, 2012, pp. 1097–1105.

[7] O. Russakovsky *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, no. 3, pp. 211–252, 2015.

[8] I. Lenz, H. Lee, and A. Saxena, "Deep learning for detecting robotic grasps," *The International Journal of Robotics Research*, vol. 34, no. 4-5, pp. 705–724, 2015.

[9] Y. Jiang, S. Moseson, and A. Saxena, "Efficient grasping from rgbd images: Learning using a new rectangle representation," in *IEEE Intl. Conf. on Robotics and Automation*, 2011, pp. 3304–3311.

[10] A. ten Pas and R. Platt, "Using geometry to detect grasp poses in 3d point clouds," in *Robotics Research*. Springer, 2018, pp. 307–324.

[11] S. Kumra and C. Kanan, "Robotic grasp detection using deep convolutional neural networks," in *Intl. Conf. on Intelligent Robots and Systems (IROS)*, 2017, pp. 769–776.

[12] C. Dong *et al.*, "Learning a deep convolutional network for image super-resolution," in *European conference on computer vision*. Springer, 2014, pp. 184–199.

[13] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *nature*, vol. 521, no. 7553, p. 436, 2015.

[14] S. El-Khoury and A. Sahbani, "A new strategy combining empirical and analytical approaches for grasping unknown 3d objects," *Robotics and Autonomous Systems*, vol. 58, no. 5, pp. 497–507, 2010.

[15] C. Borst, M. Fischer, and G. Hirzinger, "Grasp planning: How to choose a suitable task wrench space," in *IEEE Intl. Conf. on Robotics and Automation*, vol. 1, 2004, pp. 319–325.

[16] A. Saxena, J. Driemeyer, J. Kearns, C. Osondu, and A. Y. Ng, "Learning to grasp novel objects using vision," in *Experimental Robotics*. Springer, 2008, pp. 33–42.

[17] H. S. Koppula *et al.*, "Semantic labeling of 3d point clouds for indoor scenes," in *Advances in neural information processing systems*, 2011, pp. 244–252.

[18] D. Tóvári and N. Pfeifer, "Segmentation based robust interpolation-a new approach to laser data filtering," *Intl. Archives of Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 36, no. 3/19, pp. 79–84, 2005.

[19] A. Nguyen and B. Le, "Contextual labeling 3d point clouds with conditional random fields," in *Asian Conf. on Intelligent Information and Database Systems*. Springer, 2014, pp. 581–590.

[20] W. H. Chin, A. A. Saputra, Y. Toda, and N. Kubota, "An incremental episodic memory framework for topological map building," in *Intl. Electronics Symposium on Knowledge Creation and Intelligent Computing (IES-KCIC)*, 2018, pp. 322–327.

[21] W. H. Chin, A. A. Saputra, and N. Kubota, "A neuro-based network for on-line topological map building and dynamic path planning," in *Intl. Joint Conf. on Neural Networks (IJCNN)*, 2017, pp. 2805–2810.

[22] Y. Toda *et al.*, "Real-time 3d point cloud segmentation using growing neural gas with utility," in *in Proc. of 9th Intl Conf. on Human System Interactions*, 2016, pp. 418–422.

[23] R. Pelossof, A. Miller, P. Allen, and T. Jebara, "An svm learning approach to robotic grasping," in *IEEE Intl. Conf. on Robotics and Automation*, vol. 4, 2004, pp. 3512–3518.

[24] J. Redmon and A. Farhadi, "Yolov3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.